

The allegory axioms, and what defines them in the Frobenius calculus

by

§1. Cartesian bicategory of relations

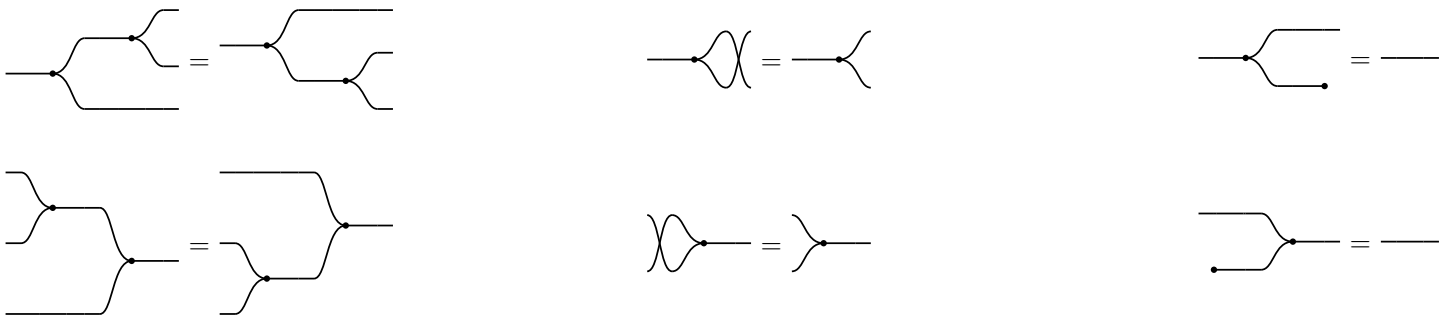
Definition 1.1.

A **cartesian bicategory of relations** is a poset-enriched category(bicategory) that is has a symmetric monoidal product $\otimes$ (cartisin product of set in Rel, vertical junctposition in the diagram) and

$\forall$ object A , $(A, \triangleleft, \multimap) \dashv (A, \triangleright, \multimap)$, **Frobenius**, arrows **lax**.

$(\triangleleft : A \rightarrow A \otimes A, \multimap : A \rightarrow \mathbb{I})$ formed a cocommutative comonoid

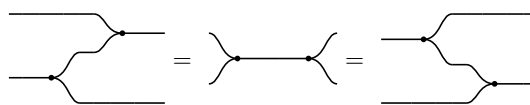
$(\triangleright : A \otimes A \rightarrow A, \multimap : \mathbb{I} \rightarrow A)$ formed a commutative monoid



§1.1. $\forall$ object A , $(A, \triangleleft, \multimap) \dashv (A, \triangleright, \multimap)$



§1.2. The Frobenius law



The only clause coupling the comonoid to the monoid beyond adjointness. Both sides reduce to the same picture, $\triangleright \triangleleft$ — merge, then copy — which is the two-in two-out **spider**. Every connected diagram built from these four generators collapses to the spider on its own inputs and outputs, and this equation is what starts that collapse. A **bubble** is the other order, $\triangleleft \triangleright$: copy then merge, a closed loop, which §4 uses.

§1.3. Every arrow is a lax comonoid homomorphism



§2. Compact closed, and where the converse comes from

Definition 2.1.

A symmetric monoidal category is **closed** when $\otimes$ can be divided by, $(- \otimes B) \dashv (B \multimap -)$ for every B , so the hom is an OBJECT of the category and not a set outside it. It is **compact** closed when every object has a dual A^* , that is a left and a right bracket satisfying the **snakes** — and then the division degenerates, $B \multimap C = B^* \otimes C$, division by B being multiplication by B^* exactly as $x / y = x y^{-1}$.

A cartesian bicategory of relations is not such a category — it is poset-enriched, and nothing in §1 asks for a dual. What it HAS is the **structure**: the two brackets are the composites $\multimap \lhd$ and $\rhd \multimap$, and the snakes below are proved, not assumed. Discard the order $\leq$ and what is left is a **hypergraph category**, also called a well-supported compact closed category, in which every object is its own dual, $A^* = A$.

In any cartesian bicategory of relations we have a self-dual compact closed structure. To describe it, we adopt the graphical notation [the two brackets, and $R^\dagger$ drawn by bending]. For an intuition consider $\mathbf{Rel}$: it is easy to check that $R^\dagger$ is just the opposite relation of R .

— Bonchi, Seeber and Sobociński

The left bracket $\multimap \lhd$ and the right bracket $\rhd \multimap$ come with the definition, and they satisfy the snakes:

$$\begin{array}{c} \text{---} \multimap \lhd \\ \text{---} \end{array} = \text{---} = \begin{array}{c} \text{---} \rhd \multimap \\ \text{---} \end{array}$$

So every object is its own dual, R° is R with its wires bent round the two brackets, and the four laws of $^\circ$ are theorems rather than assumptions:

$$(i) \ 1^\circ = 1 \quad (ii) \ (R \ S)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§3. The allegory primitives are definitions here

primitive, and what defines it here	picture
reciprocation $R^\circ := \text{bend } ((R \otimes 1) \text{ cap})$, not a generator.	
intersection $R \cap S := \lhd (R \otimes S) \rhd$ — copy, run both, merge.	
containment $R \sqsubseteq S$, which an allegory defines as $R \cap S = R$. Here $\leq$ is primitive and $\cap$ is derived, so the two directions are swapped.	the 2-cell itself
maximal morphism $\top := \multimap \multimap$, free here; an allegory needs a unit object first.	

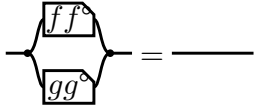
§4. The eight axioms, and what proves each

axiom, and what supplies it	picture
$(R^\circ)^\circ = R$ — the snake .	

axiom, and what supplies it	picture
$(R \circ S)^\circ = S^\circ \circ R^\circ$ — mirroring the picture.	
$(R \sqcap S)^\circ = R^\circ \sqcap S^\circ$ — the same mirroring; $\triangleleft$ and $\triangleright$ are each other's mirror image.	
$R \sqcap R = R$ — the lax copy law, then $\triangleleft \triangleright = 1$ closes the bubble.	
$R \sqcap S = S \sqcap R$ — cocommutativity and commutativity.	
$R \sqcap (S \sqcap T) = (R \sqcap S) \sqcap T$ — coassociativity and associativity.	
$R \circ (S \sqcap T) \sqsubseteq R \circ S \sqcap R \circ T$ — the lax copy law.	
$R \circ S \sqcap T \sqsubseteq (R \sqcap T \circ S^\circ) \circ S$, the modular law — adjoined as an axiom by an allegory, which has no structure to derive it from. Here Frobenius and the lax copy law give it.	

§5. Unitary and pre-tabular — what the other direction needs

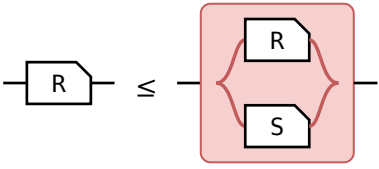
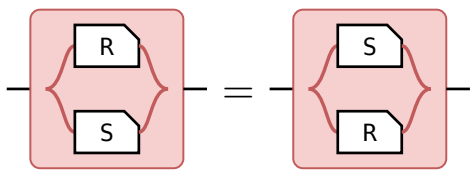
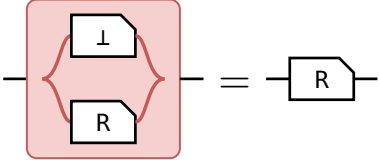
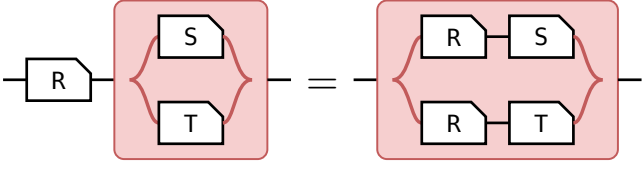
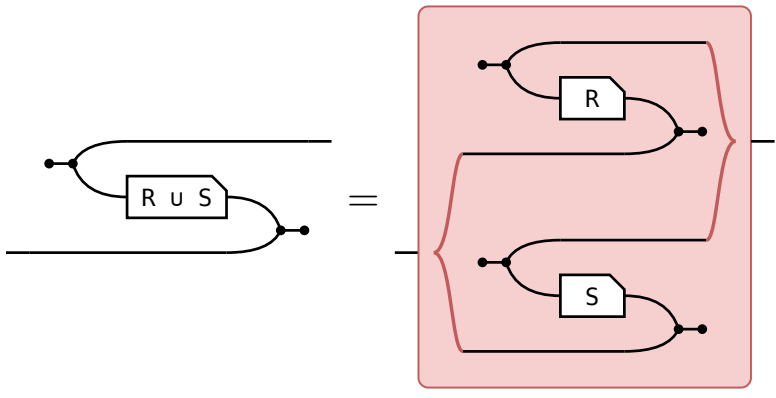
condition	allegory	Frobenius side
unitary	T is a unit when $1 \circ T$ is the largest endomorphism on it and every object carries an entire arrow to it. A unit is what makes τ exist.	free. I is the unit object, $1 \leq \multimap \multimap$ says $\multimap$ is entire, and $\tau = \multimap \multimap$ is maximal.
pre-tabular	f, g tabulate R when both are maps, $R = f^\circ g$, and	this is what $\otimes$ is. Given unitarity it reduces to tabulating each τ , and a tabulation of τ is an object $n \otimes m$ with its two projections. One side

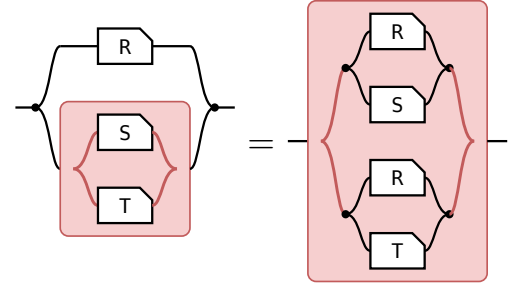
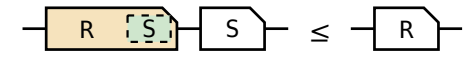
condition	allegory	Frobenius side
	 R is tabular when some pair tabulates it, the allegory pre-tabular when every arrow lies under a tabular one.	derives the product from tabulations, the other posits it.

Together: **cartesian bicategory of relations** $\simeq$ **unitary pre-tabular allegory**.

§6. Union and residual

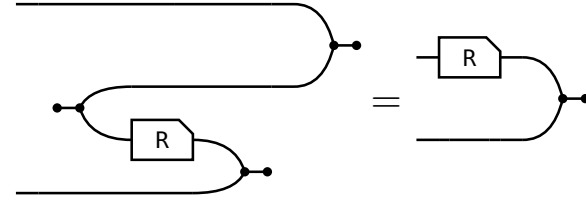
Both are past what the definition can build: a union needs a **biproduct** on top of the Frobenius structure, a residual a **second composition** with linear adjoints. A union draws as a **tape** — the rounded wrapper is the second product, and its fork and join open and close a branch a particle takes exactly one of. A residual draws as **long division**, not as its own term, which is a composition against a complement.

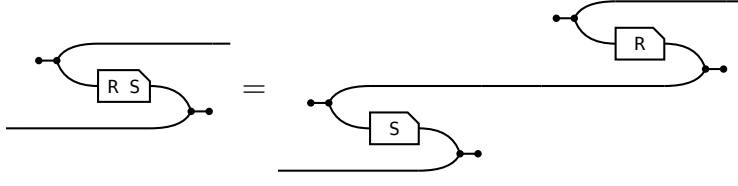
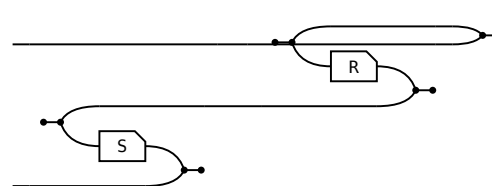
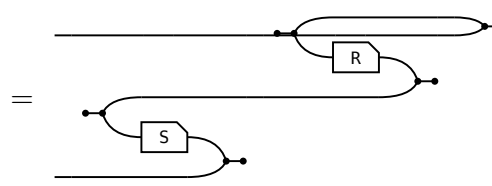
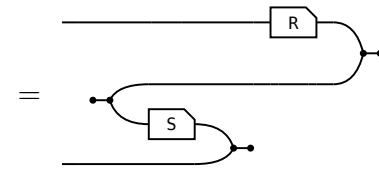
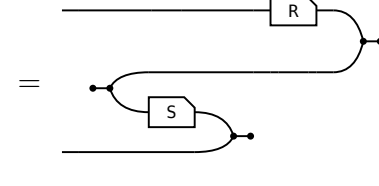
statement	picture
$R \sqsubseteq R \cup S$ union $R \ S := \langle R, S \rangle [1, 1]$: offer both branches, then forget which was taken.	
$R \cup S = S \cup R$	
$\perp \cup R = R$ $\perp$ routes through the zero object; there is no shape for it.	
$R (S \cup T) = R S \cup R T$ — composition distributes over union, which n does not.	
$(R \cup S)^\circ = R^\circ \cup S^\circ$	

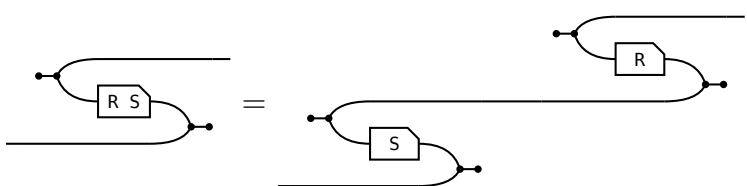
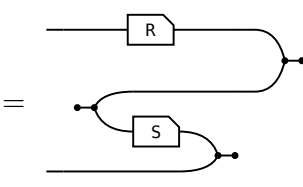
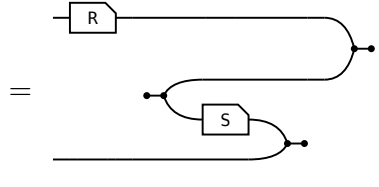
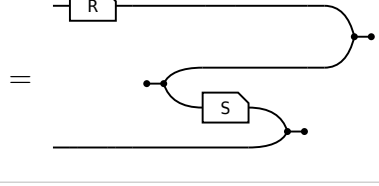
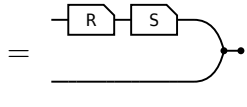
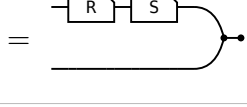
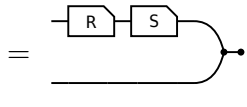
statement	picture
$R \cap (S \cup T) = (R \cap S) \cup (R \cap T)$ — what makes the whole distributive , and the one picture with both layers in it.	
$(R / S) S \sqsubseteq R$ — the residual is a lower bound of the arrows it is the greatest of.	

§7. $(R S)^\circ = S^\circ R^\circ$, drawn

Not a picture of a converse. An arrow is **the** converse of R as soon as it satisfies one equation between UNBENT arrows into $\mathbb{I}$, and bending is a bijection, so the left bracket is never drawn: every row below closes with a right bracket and opens with nothing, and “is a converse” shows up as a box on the **bottom** strand, mirrored.

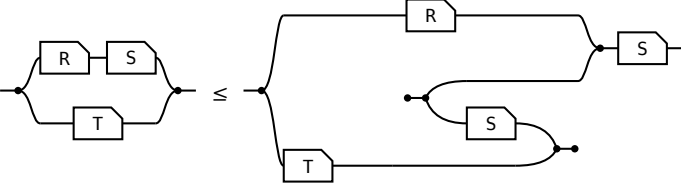
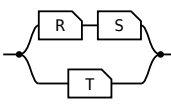
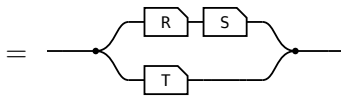
the slide — the one rule the proof uses, and it uses it twice

A box on the bottom strand facing a right bracket is that box on the top strand, upright.

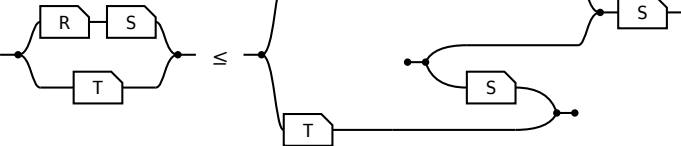
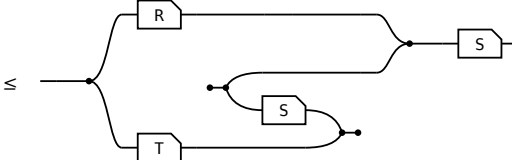
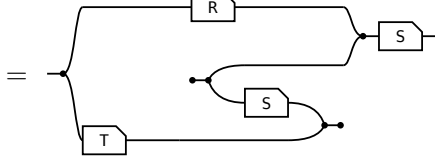
$(R S)^\circ = S^\circ R^\circ$		
term	picture	the rule that reaches it
$(1 \otimes (S^\circ R^\circ)) \text{ cap}$		the start: both boxes on the bottom strand, mirrored.
$= (1 \otimes S^\circ) ((1 \otimes R^\circ) \text{ cap})$		$\otimes$ is a functor. Same picture.
$= (1 \otimes S^\circ) ((R \otimes 1) \text{ cap})$		the slide. R° is the box nearest the right bracket: it leaves the bottom strand and comes upright as R on the top.
$= ((1 \otimes S^\circ) (R \otimes 1)) \text{ cap}$		re-bracketing. Same picture.

$(R S)^\circ = S^\circ R^\circ$ 		
term	picture	the rule that reaches it
$= (R \otimes S^\circ) \text{ cap}$		interchange. In sequence and side by side are the same thing. A re-spacing only.
$= ((R \otimes 1) (1 \otimes S^\circ)) \text{ cap}$		interchange back, R first, so S° faces the right bracket.
$= (R \otimes 1) ((1 \otimes S^\circ) \text{ cap})$		re-bracketing. Same picture.
$= (R \otimes 1) ((S \otimes 1) \text{ cap})$		the slide, on S. Both boxes now on top, upright, R then S.
$= ((R \otimes 1) (S \otimes 1)) \text{ cap}$		re-bracketing. Same picture.
$= ((R S) \otimes 1) \text{ cap}$		interchange; the two top boxes merge. This is R S unbent.

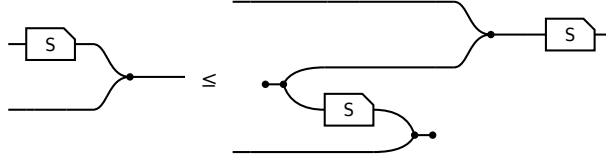
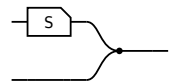
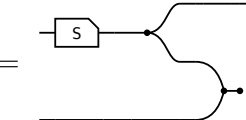
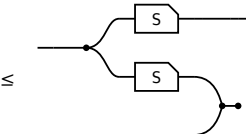
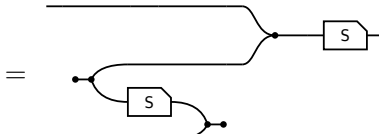
Seven of the nine steps change no picture: they are associativity, the unit laws and interchange, which a string diagram has in its geometry rather than as rewrites. The theorem is two slides.

§8. The modular law, from Frobenius

the modular law, $R S \cap T \leq (R \cap T S^\circ) S$ 		
term	picture	the rule that reaches it
$(R S) \cap T$		the start: $R S \cap T$.
$= (\triangleleft (R \otimes T)) ((S \otimes 1) \triangleright)$		Factor out the shared prefix: $(R S) \otimes T$ is $(R \otimes T) (S \otimes 1)$.

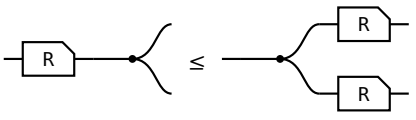
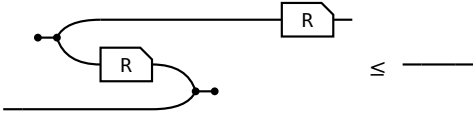
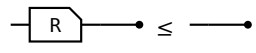
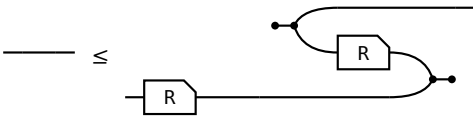
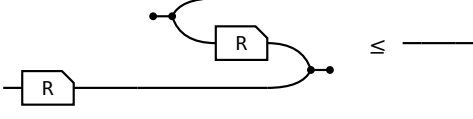
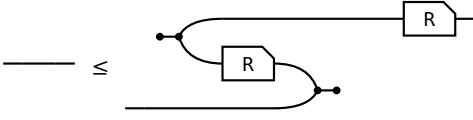
the modular law, $R \circ S \circ T \leq (R \circ T \circ S^\circ) \circ S$ 		
term	picture	the rule that reaches it
$\leq (\triangleleft (R \otimes T)) ((1 \otimes S^\circ) (\triangleright S))$		The merge slide. S slides through the merge and reappears as S° on the other strand. The only inequality in the proof.
$= (R \circ (T \circ S^\circ)) \circ S$		Fold the prefix back in; the copy-merge pair reads as a meet again.

S occurs once on the left of the slide and twice on the right, and the lax copy law is the only law in the definition that may duplicate a box — so that is where the inequality has to be.

the merge slide, $(S \otimes 1) \triangleright \leq (1 \otimes S^\circ) \triangleright S$ 		
term	picture	the rule that reaches it
$(S \otimes 1) \triangleright$		the start: $(S \otimes 1) \triangleright$.
$= ((S \triangleleft) \otimes 1) (1 \otimes \text{cap})$		Rebuild the merge from a copy and a right bracket, the shape the lax copy law applies to.
$\leq ((\triangleleft (S \otimes S)) \otimes 1) (1 \otimes \text{cap})$		The lax copy law: $S \triangleleft$ becomes $\triangleleft (S \otimes S)$, the box copied. The whole of the modular law, in one step.
$= (1 \otimes S^\circ) (\triangleright S)$		Reshape back, and the surviving duplicate slides round the right bracket to become S° .

§9. Maps

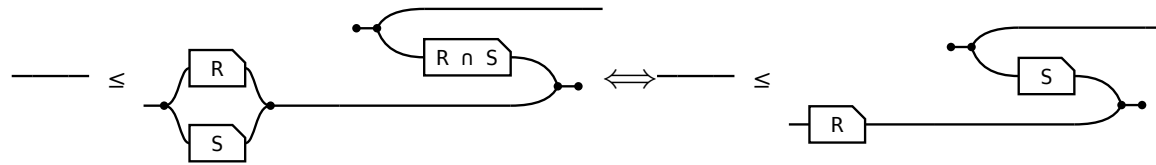
Every arrow is lax for $\triangleleft$ and for $\multimap$; (ii) and (iv) above turn those into the same statements about $\triangleright$ and $\multimap$. All four hold for **every** arrow, their REVERSES do not, and each reverse holding is a property of the arrow. The right-hand column is the **adjoint** form, which is the definition used here because it is literally the allegory's **Simple** and **Entire**; that the two forms agree is a separate theorem, not proved here.

holds for every arrow	property	holds iff
$R \triangleleft \leq \triangleleft (R \otimes R)$ 	(SV) single valued $R^\circ R \sqsubseteq 1$	
$R \multimap \leq \multimap$ 	(TOT) total $1 \sqsubseteq R R^\circ$. With single valued, a map.	
$\triangleright R \leq (R \otimes R) \triangleright$ — not stated; see below.	(INJ) injective $R R^\circ \sqsubseteq 1$	
$\circ - R \leq \circ -$ — not stated; see below.	(SUR) surjective $1 \sqsubseteq R^\circ R$, that is, R° entire.	

The two unstated ones are converse-images, and getting them needs $\triangleleft^\circ = \triangleright$ and $\multimap^\circ = \circ -$ — a Frobenius computation at a COMPOSITE object, which is the clause the printed definition omits.

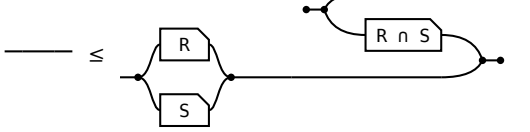
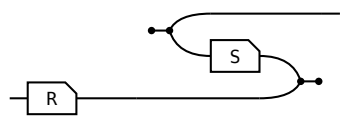
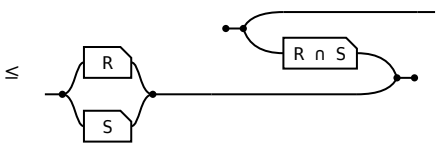
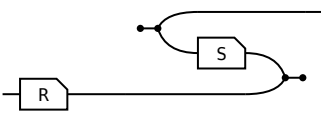
§10. Entireness of an intersection

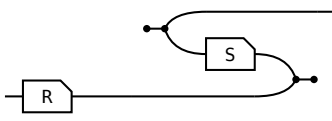
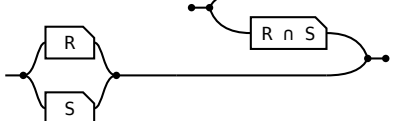
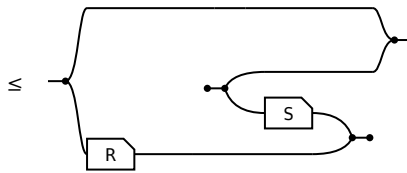
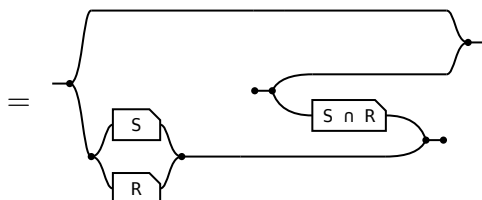
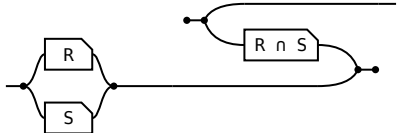
R is **entire** when $1 \sqsubseteq R R^\circ$, which is (TOT) above. When is an **intersection** entire?



$$\text{Total } (R \cap S) \leftrightarrow 1 \leq R S^\circ$$

On the left $R \cap S$ is named **twice**; on the right R and S once each and the meet is gone. The meet is needed to **ask** whether something is entire, not to **answer** it. Where $\sqsubseteq$ is **defined** as $X \cap Y = X$ the two sides are one proposition; here $\leq$ is primitive, so both directions are real steps — and they cost very different things.

$\Rightarrow$ given $R \cap S$ entire,  show 		
term	picture	the rule that reaches it
1		the start: 1.
$\leq (R \cap S) (R \cap S)^\circ$		the hypothesis: $1 \leq P P^\circ$ at $P = R \cap S$.
$\leq R S^\circ$		monotonicity, twice. No modular law.

$\Leftarrow$ given $\text{---} \leq$  $\text{show } R \cap S \text{ entire, } \text{---} \leq$ 		
term	picture	the rule that reaches it
1	---	the start: 1.
$\leq 1 \cap (R S^\circ)$		the hypothesis, met with 1.
$= 1 \cap ((S \cap R) (S \cap R)^\circ)$		$1 \cap R S^\circ = 1 \cap (S \cap R) (S \cap R)^\circ$. $R S^\circ$ relates a to a' when some b has $a R b$ and $a' S b$; the $1 \cap$ sets $a' = a$, and it reads “some b has $a R b$ and $a S b$ ” — one arrow of $R \cap S$. Two arrows become one, and that is what needs the modular law.
$\leq (R \cap S) (R \cap S)^\circ$		$X \cap Y \leq Y$, then $S \cap R = R \cap S$.

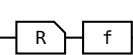
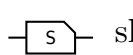
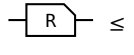
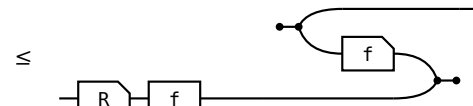
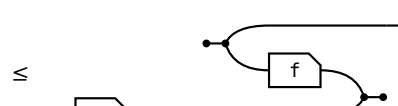
§11. The gaps

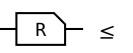
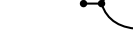
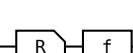
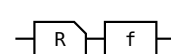
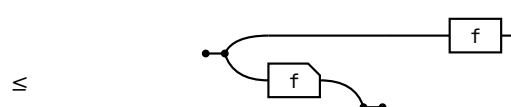
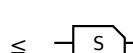
- All eight axioms are theorems of the definition, the modular identity included.
- $\triangleright \triangleleft 1$ and $\circ \circ 1$ are never used. They are there to make $\triangleleft \dashv \triangleright$ and $\circ \dashv \circ$ genuine adjunctions, which is what pins $\triangleright = \triangleleft^\circ$ and $\circ = \circ^\circ$.
- Unproved: the merge and unit forms of the lax laws, and the agreement of the comonoid and adjoint forms of the map conditions. Both trace to the one omission above.
- The bridge runs one way only: a cartesian bicategory carries $\otimes$, an allegory carries nothing of the kind, which is what the two conditions supply.
- Everything here is a theorem OF the axioms. That relations satisfy them is not checked here.

construct	counterpart	status
tabulation	none	$\tau = \circ \circ$ is all the tabular content there is.
$u, \perp$	a biproduct $\otimes$ on top of the Frobenius structure	built, and drawn above. The calculus itself has no union.
R / S , complement	a second composition with linear adjoints	built, and drawn above, but axiomatised only: nothing instantiates the complement.
Λ , power transpose	none	open. No power object anywhere in this tower.
allegory $\Rightarrow$ cartesian bicategory	rebuild $\otimes$ from tabulations	deliberately not built.

§12. The shunting rule

A map moves from one side of a containment to the other at the cost of its converse, $R f \leq S \iff R \leq S f^\circ$. It is how a function gets out of the way so the relation underneath can be reasoned about. Each direction spends exactly one half of **map** — never both.

shunting right, $\Rightarrow$ given  $\leq$  show  $\leq$ 		
term	picture	the rule that reaches it
R		the start: R.
$\leq R (f f^\circ)$		total: $1 \leq f f^\circ$, so $f f^\circ$ may be inserted after R.
$\leq S f^\circ$		Re-bracket to $(R f) f^\circ$ and apply the hypothesis.

shunting right, $\Leftarrow$ given  $\leq$  show  $\leq$ 		
term	picture	the rule that reaches it
R f		the start: R f.
$\leq S (f^\circ f)$		Apply the hypothesis on the left; the two fs are now adjacent.
$\leq S$		single valued: $f^\circ f \leq 1$, so the pair cancels.

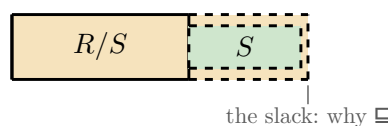
Shunting left is the mirror, same shape, map on the other side throughout. What the rule is, is an adjunction: $f \dashv f^\circ$, with **total** the unit and **single valued** the counit, so “f is a map” and “f has a right adjoint, namely f° ” are one statement.

§13. Division

Division is drawn and negation is not, because negation is not how programs get specified: the chapters where they are — greedy, thinning, dynamic programming, Horner, knapsack, paragraph formatting, string edit, bracketing, compression — use local completeness, division and power objects, and no Boolean structure at all. That also settles which picture of / is right. As a term a residual is a composition against a complement, which makes every homset Boolean; division asks for nothing of the kind, only the Galois connection $T \sqsubseteq R/S \iff T S \sqsubseteq R$, which $\mathbf{Rel}(\mathbf{Set})$ satisfies with a bare $\forall$, $(R / S) \times y \triangleq \forall z. S y z \rightarrow R x z$.

Long division. R / S is how much of R you can lay down before S still fits in what is left. Read the bar as a composite: R / S first, then S, the two together inside R. Solid is what the equation pins — R’s left edge, S’s left edge, and the stretch before S. Dashed is the slack: nothing to the right of S is determined by $T S \sqsubseteq R$, which is why the cancel law is $(R/S) S \sqsubseteq R$ and not an equality, and why the slack is a hairline — R / S is the **largest** such quotient.

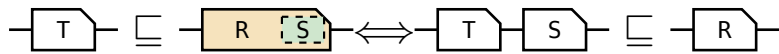
It is a **measurement**, not a containment: $R : a \rightarrow c$ and $S : b \rightarrow c$ are not in the same hom-set, so what sits inside R is the composite $(R/S) S$. Pushed too far the metaphor breaks in the usual place, $\perp / \perp = \top$.



transcribed — the universal property has a statement, this metaphor does not

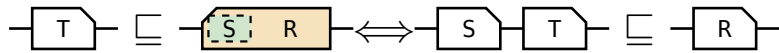
the ten laws, their pictures, and — where the metaphor reaches — the same law as long division

Division is the right adjoint of composition: $T \sqsubseteq R/S \iff T S \sqsubseteq R$



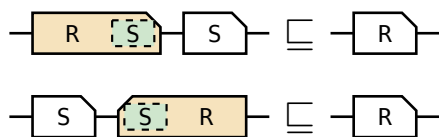
$T \sqsubseteq S \backslash R \iff S T \sqsubseteq R$

$S \backslash R \triangleq (R^\circ / S^\circ)^\circ$: left division is right division conjugated by the converse, not a second primitive.



cancel: $(R/S) S \sqsubseteq R$, and $S (S \backslash R) \sqsubseteq R$

The counit of each adjunction — the law the figure above draws.



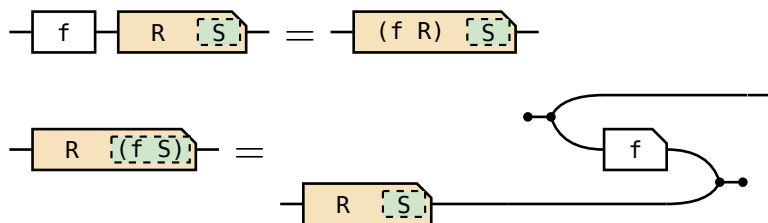
associate: $R/(S_1 S_2) = (R/S_2)/S_1$, and $(S T) \backslash R = T \backslash (S \backslash R)$

Dividing by a composite one factor at a time; the bar shows what the label on the right hides — two tiles laid where there was one.



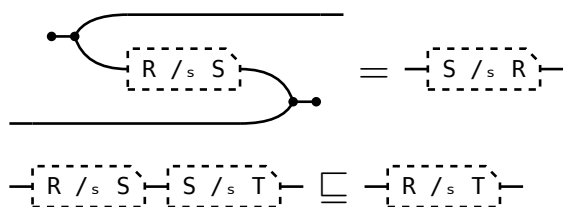
maps: $f (R/S) = (f R)/S$, and $R/(f S) = (R/S) f^\circ$

The same three pieces bracketed two ways, which is the licence to write $f R / S$; then a map moves out of a denominator and reappears as f° outside the box. Both are the shunting rule, spent twice.



symmetric division: $(R/_s S)^\circ = S/_s R$, and $(R/_s S)(S/_s T) \sqsubseteq R/_s T$

$R/_s S \triangleq (R/S) \cap (S/R)^\circ$. The meet is inside the definition, not the statement, so it stays under the label: $/_s$ is converse-symmetric and transitive, like an equality of columns.



Ten laws, ten pictures, and not one shows a generator: $\cap$, $\cup$, $^\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.